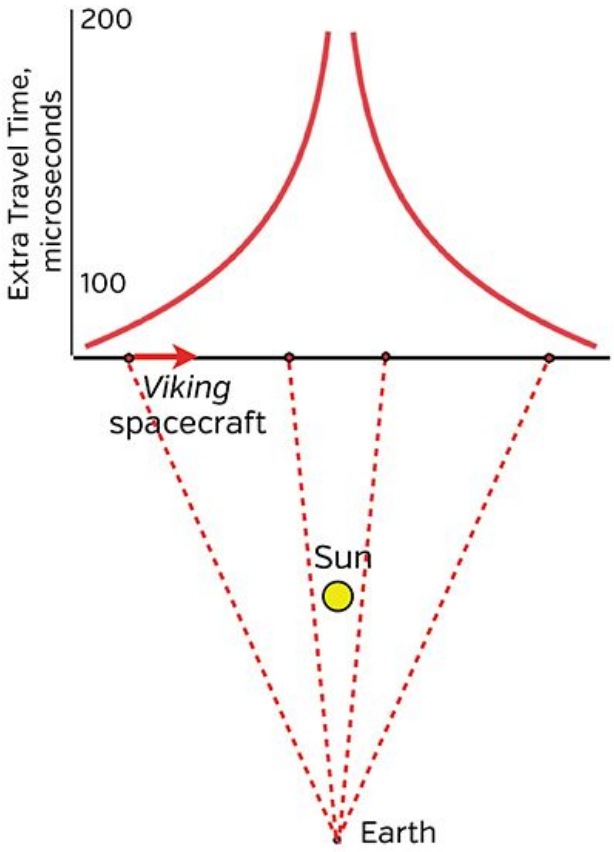


Fig. 4.3. Travel time for radio signals from Earth to *Viking* to Earth.

If space were flat, the round-trip travel time would have changed gradually and steadily. It did not. When the radio waves passed near the Sun, their travel time was longer than expected, longer by hundreds of microseconds. The extra travel time is shown, as a function of the spacecraft's location at the top of Figure 4.3; it went up and then back down. Now, one of Einstein's relativistic laws says that radio waves and light travel at an absolutely constant, unchanging speed.[10] Therefore, the distance from Earth to the spacecraft had to be longer than expected when passing near the Sun, longer by hundreds of microseconds times the speed of light: about 50 kilometers.

This greater length would be impossible if space were flat, like a sheet of paper. It is produced by the Sun's space warp. From the extra time delay and how it changed as the spacecraft moved relative to Earth,